Mediterranean Institute of Technology (Medtech)

IoT-Based Smart Agriculture System for Crop Recommendation Using Machine Learning

Introduction to Artificial Intelligence

Project Report

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1. Introduction

Modern agriculture faces significant challenges in optimizing crop production under varying environmental conditions. Farmers often rely on experience or trial-and-error when deciding which crop to plant, which can lead to poor yields or economic losses due to mismatched soil or climate conditions.

The integration of the Internet of Things (IoT) and Machine Learning (ML) offers a transformative solution to this problem. This project presents an IoT-Based Smart Agriculture System that leverages machine learning algorithms to recommend the most suitable crop based on real-time environmental and soil sensor data. By simulating an IoT environment, the system processes inputs such as soil nutrients (Nitrogen, Phosphorous, Potassium), temperature, humidity, pH, rainfall, and a computed irrigation score to intelligently predict the optimal crop.

The project was developed with the goal of applying AI concepts to a meaningful real-world problem in precision agriculture.

2. Dataset Description

The dataset used in this project is the Crop Recommendation Dataset, sourced from Kaggle. It contains 2,200 samples representing 22 different crop types, with each class having exactly 100 balanced samples. The following figure summarizes the feature definitions:

About this file Suggest Edits

N = Nitrogen
P = phosphorous
K = Potassium
Temperature=The average soil temperatures for bioactivity range from 50 to 75F.
Ph = A scale used to identify acidity or basicity nature; (Acid Nature- Ph<7; Neutral- Ph=7; Base Nature-P>7)
label = Types of Crop (Rice,Maize, Chickpea; Kidney beans; pigeonpeas; mothbeans; mungbean;blackgram; lentil; pomegranate; banana; mango; grapes; watermelon; muskmelon; apple; orange;papaya; coconut; cotton; jute; coffee)

Figure 1: Dataset feature descriptions as provided on the Kaggle dataset page.

The input features are:

- N (Nitrogen): Nitrogen content in the soil, measured in kg/ha.
- P (Phosphorous): Phosphorous content, critical for root development and energy transfer.
- K (Potassium): Potassium content, important for plant metabolism and disease resistance.
- Temperature: Average ambient/soil temperature suitable for bioactivity (50–75°F range).
- Humidity: Relative humidity percentage of the surrounding environment.
- Ph: Soil pH scale — Acid (Ph < 7), Neutral (Ph = 7), Basic (Ph > 7).
- Rainfall: Annual rainfall in millimeters.
- Label (Target): The recommended crop type — one of 22 classes including Rice, Maize, Chickpea, Kidney Beans, Pigeonpeas, Mothbeans, Mungbean, Blackgram, Lentil, Pomegranate,

Banana, Mango, Grapes, Watermelon, Muskmelon, Apple, Orange, Papaya, Coconut, Cotton, Jute, and Coffee.

The perfectly balanced class distribution (100 samples per crop) is a critical property that ensures no model bias toward any particular crop during training.

3. Methodology

The project follows a standard supervised machine learning pipeline, from raw data ingestion through model training to final crop prediction. Each stage is described in detail below.

3.1 Data Loading and Exploration

The dataset was loaded from a CSV file using the Pandas library. An initial exploration was conducted to inspect the structure of the dataset: the number of records and columns, data types, basic statistical summaries (mean, standard deviation, minimum and maximum values for each feature), and the distribution of crop labels. This step confirmed that the dataset was clean, complete, and class-balanced — an ideal foundation for training classification models.

3.2 Data Visualization

Two visualizations were produced to better understand the data and assess its suitability for machine learning:

Crop Distribution Chart

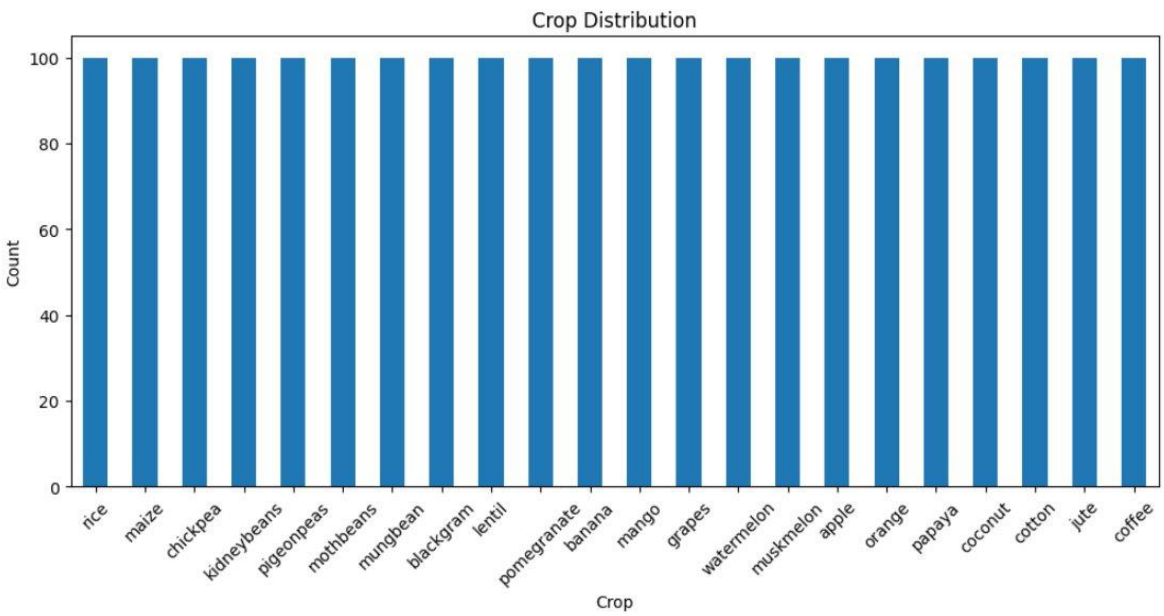

Figure 2: Crop Distribution — Each of the 22 crop classes contains exactly 100 samples, confirming a perfectly balanced dataset.

The bar chart confirms that every crop class is represented by exactly 100 samples. This balance is essential because machine learning models trained on imbalanced data tend to be biased toward majority classes. With a balanced dataset, no additional resampling techniques (such as oversampling or SMOTE) were required, and evaluation metrics such as accuracy are reliable indicators of true model performance.

Principal Component Analysis (PCA) Visualization

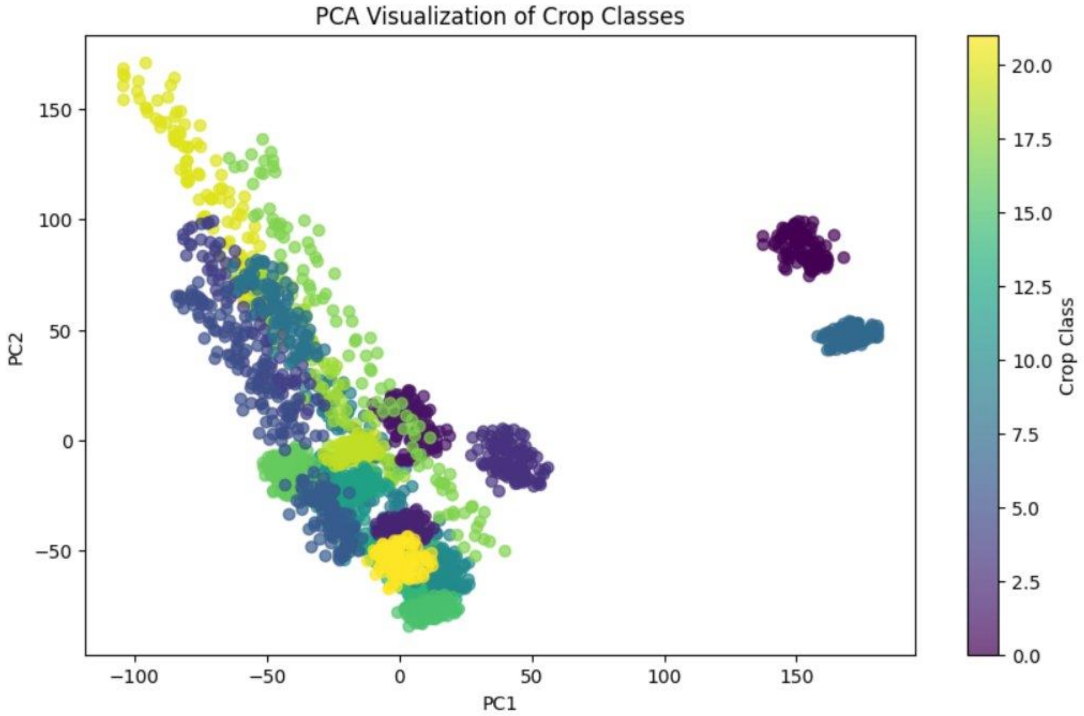

Figure 3: PCA Visualization — 2D projection of all 2,200 crop samples. Distinct cluster structures indicate learnable patterns in the data.

Principal Component Analysis (PCA) is a dimensionality reduction technique that projects high-dimensional data into fewer dimensions while retaining maximum variance. The 7 features were reduced to 2 principal components for visualization. The scatter plot reveals meaningful structure: most crops form an overlapping diagonal cluster, while two isolated groups (visible on the far right, approximately at PC1 = 150–180) correspond to crops with highly distinct environmental profiles, most likely Grapes and Apple. The presence of this clear structure confirms that the dataset contains learnable patterns, validating the use of supervised ML classification.

3.3 Feature Engineering

To enrich the dataset with agronomic domain knowledge, a new synthetic feature — the Irrigation Score — was computed from the existing sensor readings. This integer score (ranging from 0 to 4) quantifies how much supplemental irrigation a given field is likely to require, based on the following rules:

- Humidity below 60%: +1 point (dry air conditions increase evapotranspiration).
- Rainfall below 100 mm: +1 point (low natural water supply requires supplemental irrigation).

- Temperature above 30°C: +1 point (high temperatures increase water loss through evaporation).
- Soil pH outside 5.5–7.5: +1 point (extreme pH values impair nutrient and water absorption).

This engineered feature simulates the kind of derived metric that a real IoT gateway would compute from raw sensor readings. It adds contextual agricultural intelligence to the raw measurements and, as confirmed by the feature importance analysis, contributes meaningfully to the final model's predictions.

3.4 Data Preprocessing

After feature engineering, the following preprocessing steps were applied in sequence:

- Feature/Target Separation: The dataset was split into the feature matrix X (all input columns) and the target vector y (the crop label).
- Label Encoding: Categorical crop names in y were converted to integers (e.g., "rice" → 0, "maize" → 1) to make them compatible with scikit-learn models.
- Train/Test Split: The data was divided into 80% training (1,760 samples) and 20% test (440 samples) sets using a fixed random seed (42) to ensure reproducibility.
- Feature Scaling (StandardScaler): All features were standardized to zero mean and unit variance using the formula X_scaled = (X − mean) / std. This step is critical for Logistic Regression, which is sensitive to feature magnitudes, and ensures a fair comparison across all three models.

4. Machine Learning Models

Three supervised classification models were trained and evaluated on the same preprocessed dataset. Each model was selected to represent a distinct class of algorithm, allowing a systematic comparison from linear to non-linear to ensemble approaches.

4.1 Model 1: Logistic Regression

Logistic Regression is a foundational linear classification algorithm. Despite its name, it is used for classification: it learns a linear decision boundary in the feature space and applies the sigmoid function to output class probabilities. It was included in this study as a baseline model for the following reasons:

- It is highly interpretable and computationally efficient.
- It establishes a linear performance benchmark against which non-linear models can be fairly compared.
- The PCA visualization suggested that some degree of linear separability exists in the dataset, making Logistic Regression a reasonable first choice.

The model was trained with a maximum of 1,000 iterations to ensure convergence of the optimization solver. It achieved an accuracy of approximately 97% on the test set.

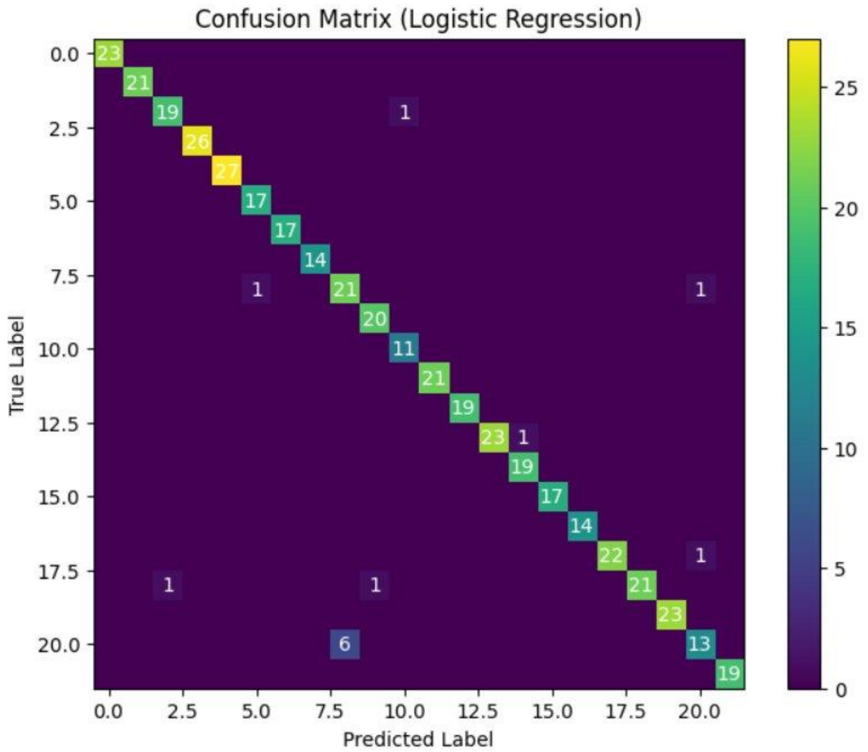

Figure 4: Confusion Matrix — Logistic Regression (~97% accuracy). Most predictions are correct (on the diagonal); minor off-diagonal entries reflect misclassifications between crops with similar environmental profiles.

The confusion matrix shows that Logistic Regression correctly classifies the vast majority of crop samples. A small number of off-diagonal entries are visible, indicating that a linear boundary is insufficient to perfectly separate crops with highly overlapping environmental conditions — a finding consistent with the PCA scatter plot. Nevertheless, 97% accuracy on a 22-class problem is a strong result for a linear model.

4.2 Model 2: Decision Tree

A Decision Tree is a non-linear, tree-structured classifier that learns a hierarchical sequence of if-else conditions on the input features to partition data into leaf nodes, each associated with a crop class. It was included because:

- It can capture non-linear and complex relationships that a linear model like Logistic Regression cannot.
- It is fully interpretable — the learned tree can be visualized and explained to agronomists without any machine learning background.
- It serves as the fundamental building block for the Random Forest ensemble, making it important to understand individually before scaling it up.

The Decision Tree was grown without a depth limit, allowing it to keep splitting until all training samples were correctly classified. It achieved approximately 99% accuracy on the test set.

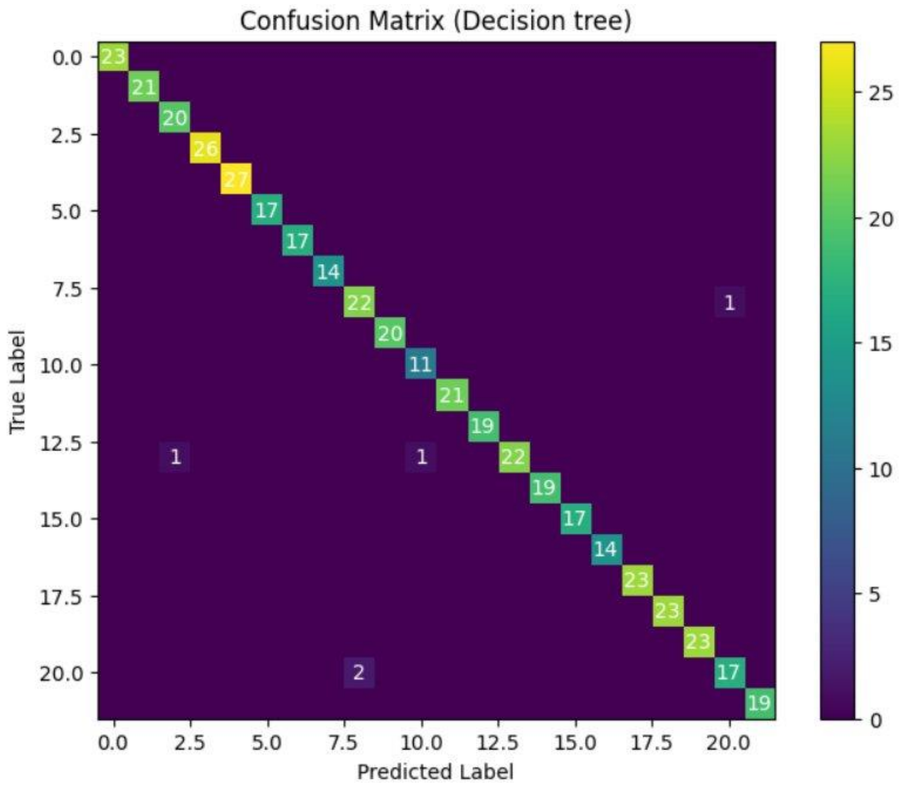

Figure 5: Confusion Matrix — Decision Tree (~99% accuracy). Nearly all predictions are correct, with only a handful of misclassifications.

While the Decision Tree achieved higher accuracy than Logistic Regression, the absence of depth pruning raises concerns about overfitting. An unconstrained tree can memorize training data — including noise — by creating excessively deep branches, which may reduce generalization to truly unseen field data. This limitation motivated the adoption of an ensemble approach.

4.3 Model 3: Random Forest

Random Forest is an ensemble learning method that constructs a collection of 100 decision trees during training and outputs the class receiving the majority vote across all trees. Each tree is trained on a randomly sampled bootstrap subset of the training data (bagging) and uses a random subset of features at each split. This double injection of randomness serves to decorrelate the individual trees, dramatically reducing variance and overfitting. Random Forest was chosen as the primary model of interest because:

- It directly addresses the overfitting problem of individual Decision Trees by averaging predictions across 100 trees.
- It produces robust, generalizable predictions even on noisy or real-world sensor data.
- It natively computes feature importances, offering agronomic interpretability alongside high accuracy.
- It consistently achieves state-of-the-art performance on structured/tabular datasets in the literature.

The Random Forest was configured with 100 estimators and a fixed random seed (42). It achieved the highest accuracy of approximately 99–100% on the test set.

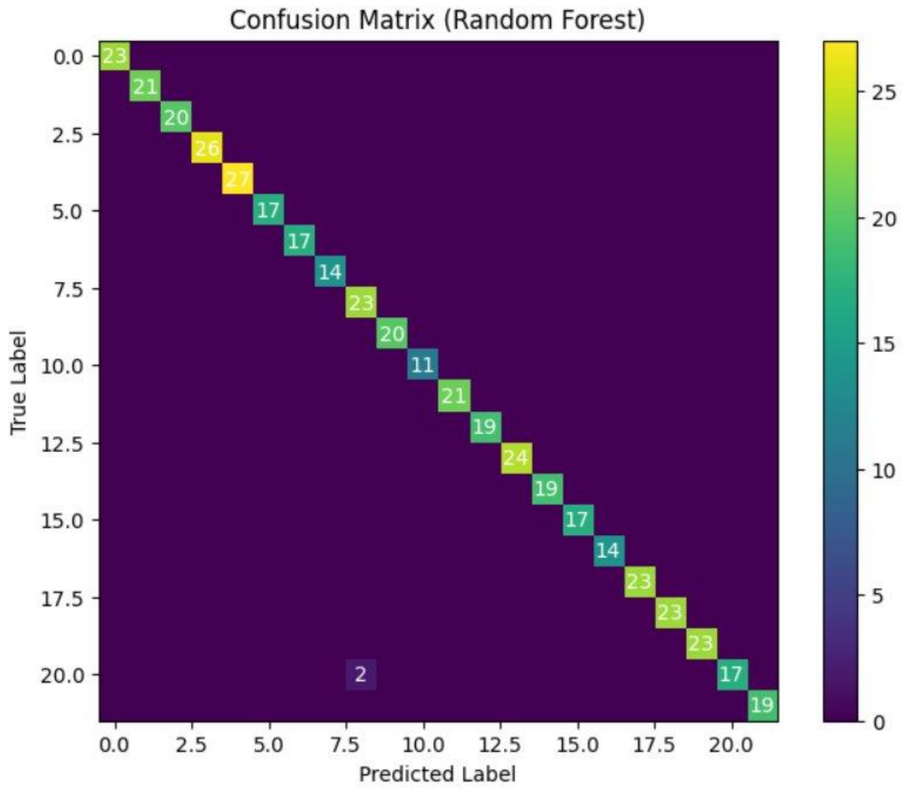

Figure 6: Confusion Matrix — Random Forest (~99–100% accuracy). The near-perfect diagonal demonstrates exceptional classification performance across all 22 crop types.

The Random Forest confusion matrix is nearly perfect. By combining 100 decision trees, the ensemble effectively cancels out individual tree errors, yielding a highly reliable crop recommendation system. This result validates the choice of Random Forest as the final deployed model.

5. Model Comparison and Final Selection

The three models were evaluated on the same held-out test set (20% of the dataset). The table below summarizes the results:

Model	Test Accuracy	Remarks
Logistic Regression	~97%	Linear baseline; slight misclassifications on overlapping crops
Decision Tree	~99%	Non-linear; risk of overfitting without depth pruning
Random Forest ✓	**~99–100%**	**Ensemble; best generalization — SELECTED as final model**

Random Forest was selected as the final model for the following reasons:

- Superior accuracy: Achieved the highest test accuracy, outperforming both baseline models.
- Reduced overfitting: Ensemble averaging prevents the memorization of training noise that affects single Decision Trees, resulting in better generalization to new IoT sensor readings.
- Feature interpretability: Random Forest provides quantified feature importances, which is invaluable for helping farmers and agronomists understand which soil and climate factors matter most.
- Real-world robustness: Random Forest is inherently robust to sensor outliers and mild noise, making it well-suited for deployment in real IoT agricultural environments.

6. Feature Importance Analysis

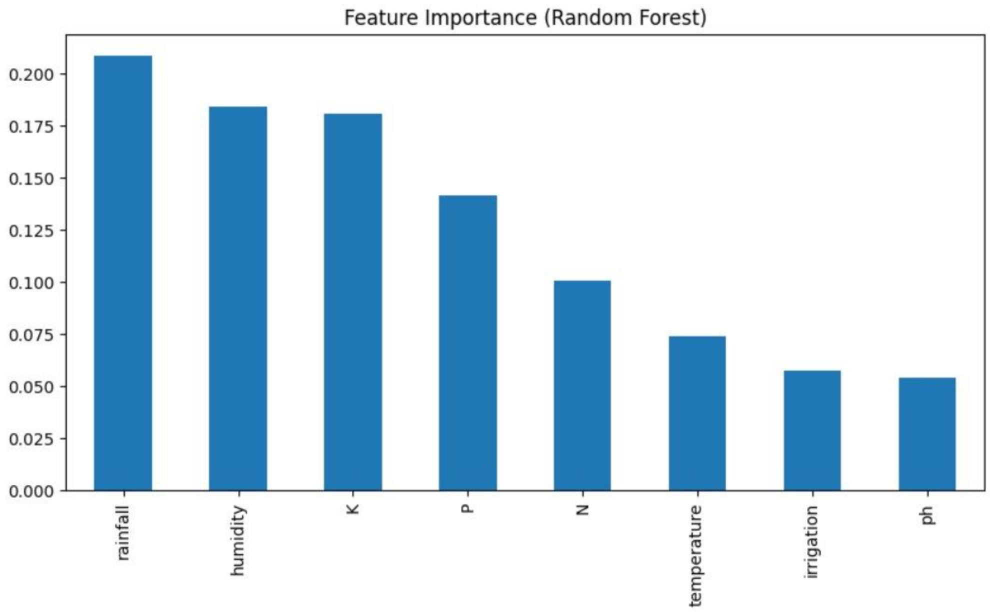

Figure 7: Feature Importance — Random Forest. Rainfall is the single most influential predictor, followed closely by humidity and potassium.

The Random Forest model ranks each input feature by the average reduction in node impurity (Gini importance) it produces across all 100 trees. The results reveal the following hierarchy:

- Rainfall (~0.21): The most influential feature. Water availability is the primary determinant of which crops can survive in a given region — a finding consistent with agronomic science.
- Humidity (~0.18): Closely following rainfall, ambient humidity levels differentiate crops adapted to tropical versus arid climates.
- Potassium / K (~0.18): Critical for plant metabolism, photosynthesis, and immune response; varies substantially across crop types.
- Phosphorous / P (~0.14): Important for root development and energy transfer within plants.
- Nitrogen / N (~0.10): Drives leaf and stem growth; its variation distinguishes leguminous crops (which fix atmospheric nitrogen) from cereals.
- Temperature (~0.075): Influences growing seasons and determines which crop varieties are climatically viable.

- Irrigation Score (~0.056): The engineered feature delivers meaningful predictive power, confirming that the feature engineering step added genuine value to the model.
- pH (~0.054): Soil acidity controls nutrient bioavailability and is a key differentiator for acid-sensitive crops such as blueberries or grapes.

Notably, feature importance is relatively evenly distributed across all eight features. No single feature dominates to the extent that others become redundant, confirming that a multi-sensor IoT platform collecting all these parameters is necessary and justified for accurate crop recommendation.

7. Inference on New Data

To validate the practical utility of the trained system, the Random Forest model was tested on a hypothetical new field reading simulating IoT sensor output:

- Soil nutrients: N = 90 kg/ha, P = 40 kg/ha, K = 40 kg/ha
- Environmental conditions: Temperature = 25°C, Humidity = 80%, pH = 6.5, Rainfall = 200 mm
- Computed Irrigation Score: 1 (low irrigation need — high humidity and high rainfall partially offset the other conditions)

The new sample was passed through the same StandardScaler fitted on the training data, ensuring consistent feature normalization. The Random Forest model then produced a numerical prediction, which was decoded back to a human-readable crop name using the inverse Label Encoder. The model predicted Rice as the most suitable crop for these conditions — a scientifically sound result, as rice is known to thrive in high-humidity, high-rainfall environments with slightly acidic soil.

This end-to-end demonstration shows the full pipeline: from raw sensor data collection to a concrete, actionable crop recommendation. Such a system could realistically be deployed on an embedded IoT gateway (e.g., a Raspberry Pi connected to field sensors) to provide real-time crop guidance to farmers.

8. Conclusion

This project successfully developed an end-to-end IoT-based smart agriculture system for crop recommendation using machine learning. Three classification models — Logistic Regression (~97%), Decision Tree (~99%), and Random Forest (~99–100%) — were trained, evaluated, and compared on the Crop Recommendation Dataset.

Random Forest was selected as the final model due to its superior performance, robustness against overfitting, and built-in interpretability through feature importances. The analysis identified rainfall, humidity, and potassium as the three most critical environmental factors influencing crop selection. A custom irrigation score feature was successfully engineered from raw sensor inputs, further enhancing prediction quality.

The project demonstrates that combining IoT sensor infrastructure with supervised machine learning is a feasible and effective approach to precision agriculture. Future extensions of this work could include: deployment on real embedded hardware with physical soil and climate sensors; integration of real-time weather API data for dynamic recommendations; expansion of the dataset to cover more crop varieties and geographic regions; and exploration of advanced deep learning models for further accuracy improvements.

9. References

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